TOR INDEPENDENCE PRESERVES TOR AMPLITUDE AND DETECTS FLATNESS UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Tor dimension of objects of the derived category can be computed at stalks, by the cohomology tor amplitude stalk lemma. invoked at proof lines 5533-5534. Reduces part (1) to a stalkwise computation.
- For a tor independent cartesian square the derived pullback stalk identifies with the derived tensor product over the base ring, by the More on Algebra base change comparison lemma. invoked at proof lines 5542-5545. Rewrites the stalk of L(g')^*E as a tensor over O_{S,s}.
- For modules over rings, tor amplitude is preserved under derived tensor with a ring of tor amplitude in degree zero, by the More on Algebra pull tor amplitude lemma. invoked at proof lines 5546-5547. Concludes part (1); part (2) follows by reading flatness as tor amplitude in [0, 0].

Source reference. Original lemma 09U9: sources/stacks-chow/tags.txt, source lines 12417–12417. Cited at proof lines 5533-5534 to reduce to stalks; the stalk rewriting is inside the counted span.

Source reference. Original lemma 0661: sources/stacks-chow/tags.txt, source lines 7920–7920. Cited at proof lines 5543-5545 to rewrite the stalk over the base ring; the tor amplitude conclusion is inside the counted span.

Source reference. Original lemma 066L: sources/stacks-chow/tags.txt, source lines 7940–7940. Cited at proof lines 5546-5547 to conclude part (1); part (2) is inside the counted span.

Lemma 0.2. *Consider a cartesian square of schemes*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

Assume g and f Tor independent.

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(1) *If $E \in D(\mathcal{O}_X)$ has tor amplitude in $[a, b]$ as a complex of $f^{-1}\mathcal{O}_S$-modules, then $L(g')^*E$ has tor amplitude in $[a, b]$ as a complex of $f^{-1}\mathcal{O}_{S'}$-modules.*
(2) *If $\mathcal{G}$ is an $\mathcal{O}_X$-module flat over S, then $L(g')^*\mathcal{G} = (g')^*\mathcal{G}$.*

Proof. We can compute tor dimension at stalks, see Cohomology, Lemma 09U9. If $x' \in X'$ with image $x \in X$, then

$$(L(g')^*E)_{x'} = E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} \mathcal{O}_{X',x'}$$

Let $s' \in S'$ and $s \in S$ be the image of x' and x. Since X and S' are tor independent over S, we can apply More on Algebra, Lemma 0661 to see that the right hand side of the displayed formula is equal to $E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{S,s}} \mathcal{O}_{S',s'}$ in $D(\mathcal{O}_{S',s'})$. Thus (1) follows from More on Algebra, Lemma 066L. To see (2) observe that flatness of $\mathcal{G}$ is equivalent to the condition that $\mathcal{G}[0]$ has tor amplitude in $[0, 0]$. Applying (1) we conclude. □